5V_IN
VIN_RAW
GND
F1
0.5A hold
VIN_FUSED
Q1_GATE
AO3401A
Q1
10k
R1
D1
SMAJ6.0A
10uF
C1
100nF
C2
C5
U1
AMS1117LV33
+3V3_P
C3
C4
1.5k
R2
LED1_A
RED 5V
LED1
680
R3
LED2_A
GREEN 3V3
LED2
3V3_ESP
ESP32_IF
3V3_ESP
GND
US_TRIG_ESP
US_ECHO_ESP
I2C_SDA
I2C_SCL
UART_TX_ESP
UART_RX_ESP
GP_IO1_ESP
GP_IO2_ESP
J2
U3
SN74LVC1T45DBVR
VCCA
VCCB
DIR
US_TRIG_5V
100k
R4
R5
C6
C7
ULTRASONIC
US_ECHO_5V
C8
C9
R6
US_ECHO_IN
R7
U2
SN74LVC1G17DBVR
C10
ECHO_ESP
TRIG_5V
I2C
J5
I2C_PU
2.2k
R8
R9
I2C_PU_EN
JP2
UART
UART_VCC
UART_TX
UART_RX
JP1
UART_VCC_SEL
UART_TX_ESP
UART_RX
UART_RX_IN
220
R10
1k
R11
100R
R12
U4
UART_RX_IN
UART_RX_ESP
C11
GP_3V3
J6
GP_IO1
GP_IO2
R13
R14
PWR_FLAG

Architecture B: ESP32 DevKit on its own USB; board powers peripherals
AI-PCB-Engineer

Sheet: /
File: Sensor_Interface_Power_Board.kicad_sch

Title: Sensor Interface & Power Management Board for ESP32

Size: A3	Date: 2026-11-08	**Rev: 0.1.0**
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